As the above is an application, compile it with *gcc part_info.c -o part_info*, and then run *./part_info /dev/sda* to check out your primary partitioning information on */dev/sda*. Figure 2 shows the output of *./part_info* on the presenter's system. Compare it with the *fdisk* output in Figure 1.

Partition types and boot records

Now, as this partition table is hard-coded to have four entries, that's the maximum number of partitions you can have. These are called primary partitions, each having an associated type in the corresponding partition table entry. These types are typically coined by various OS vendors, and hence sort of map to various OSs like DOS, Minix, Linux, Solaris, BSD, FreeBSD, QNX, W95, Novell Netware, etc, to be used for/with the particular OS. However, this is more a formality than a real requirement.

Besides this, one of the four primary partitions can be labelled as something called an *extended partition*, which has a special significance. As the name suggests, it is used to further extend hard disk division, i.e., to have more partitions. These are called *logical partitions* and are created within the extended partition. The metadata of these is maintained in a linked-list format, allowing an unlimited number of logical partitions (at least theoretically). For that, the first sector of the extended partition, commonly called the Boot Record (BR), is used like the MBR to store (the linked-list head of) the partition table for the logical partitions. Subsequent linked-list nodes are stored in the first sector of the subsequent logical partitions, referred to as the Logical Boot Record (LBR). Each linked-list node is a complete 4-entry partition table, though only the first two entries are used—the first for the linked-list data, namely, information about the immediate logical partition, and the second as the linked list's next pointer, pointing to the list of remaining logical partitions.

To compare and understand the primary partitioning details on your system's hard disk, follow the steps (as the root user—hence with care) given below:

- *./part_info /dev/sda* # Displays the partition table on */dev/sda*
- *fdisk -l /dev/sda* # To display and compare the partition table entries with the above

In case you have multiple hard disks (*/dev/sdb*, ...), hard disk device files with other names (*/dev/hda*, ...), or an extended partition, you may try *./part_info <device_file_name>* on them as well. Trying on an extended partition would give you the information about the starting partition table of the logical partitions.

Right now, we have carefully and selectively played (read-only) with the system's hard disk. Why carefully? Since otherwise, we may render our system non-bootable. But no learning is complete without a total exploration. Hence, in our next session, we will create a dummy disk in RAM and do destructive exploration on it. END

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